Hence, for all $n > 198$:

$$\begin{aligned} a_n &= \frac{n}{n} \cdot \frac{n-1}{n} \cdots \frac{n-98}{n} \cdot \frac{n-99}{n} \cdot (n-100)! \\ &> \frac{1}{2} \cdot \frac{1}{2} \cdots \frac{1}{2} \cdot \frac{1}{2} \cdot (n-100)! \\ &= \frac{1}{2^{100}} \cdot (n-100)! \end{aligned}$$

Now, $\frac{1}{2^{100}}$ is some (*very* small) positive number, however $(n-100)!$ diverges as $n \to \infty$, so that eventually, for any $A > 0$, there exists $N_A \in \mathbb{N}$ such that $\frac{1}{2^{100}} \cdot (n-100)! > A$ for all $n > N_A$. Hence

$$\lim_{n\to\infty} a_n = +\infty.$$

Indeterminate sequences

Some sequences do not converge but also do not diverge. These are called **indeterminate sequences**. Examples include:

- The sequence $a_n = (-1)^n$ is *bounded* but does *not* converge.
- The sequence $a_n = n(-1)^n$ is *not* bounded (not from below and not from above) *and* does *not* converge.

A crucial feature of indeterminate sequences is that their values are not monotone: they increase and decrease repeatedly. The following theorem demonstrates this.

Theorem 5.1 (Monotone Convergence Theorem)**:** Let $\{a_n\}_{n\in\mathbb{N}}$ be a real-valued sequence. Assume that there exists $N \in \mathbb{N}$ such that for all $n > N$, the sequence is monotone. Then the sequence *cannot* be indeterminate. More precisely:

- In the case that the sequence is monotone increasing:
  - If $\sup\{a_n \mid n > N\} < +\infty$, then $\lim_{n\to\infty} a_n = \sup\{a_n \mid n > N\}$.
  - If $\sup\{a_n \mid n > N\} = +\infty$, then the sequence diverges to $+\infty$.
- In the case that the sequence is monotone decreasing:
  - If $\inf\{a_n \mid n > N\} > -\infty$, then $\lim_{n\to\infty} a_n = \inf\{a_n \mid n > N\}$.
  - If $\inf\{a_n \mid n > N\} = -\infty$, then the sequence diverges to $-\infty$.

Proof. We only prove for the monotone *increasing* case (the decreasing case follows the same proof). For brevity we shall write

$$\sup_{n>N} a_n = \sup\{a_n \mid n > N\}$$

$\star$ Suppose that $\sup_{n>N} a_n = \ell < +\infty$. Fix some $\varepsilon > 0$. By the definition of the supremum,

1. there exists some index $N_\varepsilon > N$ such that $\ell - a_{N_\varepsilon} < \varepsilon$;
2. for all $n > N$, $a_n \leq \ell$.

Combining these with the fact that the sequence is monotone increasing for $n > N$, we have the following sequence of inequalities

$$a_{N+1} \leq a_{N+2} \leq \cdots \leq \underbrace{a_{N_\varepsilon}}_{>\ell-\varepsilon} \leq a_{N_\varepsilon+1} \leq \cdots \leq \ell$$

Neglecting the terms up to a_{N_ε}, this can be written as

$$\ell - \varepsilon < a_{N_\varepsilon} \leq a_{N_\varepsilon+1} \leq \cdots \leq \ell$$

This means that for all $n \geq N_\varepsilon$, $|\ell - a_n| < \varepsilon$. By the definition of the limit of a sequence, this means that

$$\lim_{n\to\infty} a_n = \ell.$$

$\star$ Now, suppose that $\sup_{n>N} a_n = +\infty$. Then (by definition) for every $A > 0$, there exists $N_A > N$ such that $a_{N_A} > A$. So we have

$$A < a_{N_A} \leq a_{N_A+1} \leq \cdots$$

By definition, this precisely means that

$$\lim_{n\to\infty} a_n = +\infty.$$

□

Example 5.3:
1. The sequence $a_n = \frac{n}{n+1}$ $(n \in \mathbb{N})$ is monotonically increasing, and its supremum is 1. Hence its limit exists (and it is 1).
2. The sequence $a_n = \frac{1}{n}$ $(n \in \mathbb{N}_+)$ is monotonically decreasing and its infimum is 0. Hence its limit exists, and it is also 0.

Proposition 5.2 (The number e)**:** The sequence $a_n = \left(1 + \frac{1}{n}\right)^n$, $n \in \mathbb{N}$, is monotonically increasing and bounded from above. Hence it has a limit, which is denoted e (this is the famous Euler's number, and this is how it is defined):

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n.$$

Proof. We need to show that the sequence a_n is bounded and monotonically increasing.

The sequence is monotonically increasing. Actually, it is strictly increasing. We write

$$\begin{aligned} a_n = \left(1 + \frac{1}{n}\right)^n &= \sum_{k=0}^{n} \binom{n}{k} \frac{1}{n^k} = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} \frac{1}{n^k} = \sum_{k=0}^{n} \frac{1}{k!} \frac{n(n-1)\cdots(n-k+1)}{n^k} \\ &= \sum_{k=0}^{n} \frac{1}{k!} \frac{n}{n} \cdot \frac{n-1}{n} \cdots \frac{n-k+1}{n} \\ &= \sum_{k=0}^{n} \frac{1}{k!} 1 \cdot \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right). \end{aligned}$$

Similarly, we can express a_{n+1} as:

$$a_{n+1} = \sum_{k=0}^{n+1} \frac{1}{k!} 1 \cdot \left(1 - \frac{1}{n+1}\right) \cdots \left(1 - \frac{k-1}{n+1}\right).$$

Comparing a_n and a_{n+1} we see that:

$$\begin{aligned} a_n &= \sum_{k=0}^{n} \frac{1}{k!} 1 \cdot \left(1 - \frac{1}{n}\right) \cdots\cdots \left(1 - \frac{k-1}{n}\right) \\ &\qquad\qquad \wedge \qquad\qquad\qquad \wedge \\ a_{n+1} &= \sum_{k=0}^{n+1} \frac{1}{k!} 1 \cdot \left(1 - \frac{1}{n+1}\right) \cdots \left(1 - \frac{k-1}{n+1}\right) \end{aligned}$$

each term in the product is bigger in the expression for a_{n+1} (and, moreover, a_{n+1} has an additional positive summand $k = n+1$). Therefore $a_{n+1} > a_n$ (strict inequality).

The sequence is bounded. Observe that $a_1 = 2$, so that 2 is a lower bound (the sequence is increasing). We will now show that 3 is an upper bound. We shall use the inequality

$$k! = \underbrace{k(k-1)(k-2)\cdots 2}_{k-1 \text{ terms}} \cdot 1 \geq \underbrace{2 \cdot 2 \cdots 2}_{k-1 \text{ times}} = 2^{k-1}.$$

We write a_n as before:

$$\begin{aligned} a_n &= \sum_{k=0}^{n} \frac{1}{k!} 1 \cdot \left(1 - \frac{1}{n}\right) \cdots\cdots \left(1 - \frac{k-1}{n}\right) \\ &< \sum_{k=0}^{n} \frac{1}{k!} = 1 + \sum_{k=1}^{n} \frac{1}{k!} \\ &\leq 1 + \sum_{k=1}^{n} \frac{1}{2^{k-1}} \\ &= 1 + \sum_{k=0}^{n-1} \frac{1}{2^k}. \end{aligned}$$

We know the formula for the partial sum of a geometric series:

$$\sum_{k=0}^{n-1} \frac{1}{2^k} = \frac{1 - \frac{1}{2^n}}{1 - \frac{1}{2}} = 2\left(1 - \frac{1}{2^n}\right) < 2.$$

So we find that

$$a_n < 1 + \sum_{k=0}^{n-1} \frac{1}{2^k} < 1 + 2 = 3.$$

□

> **Leonhard Euler (1707–1783)**
>
> *Euler is the mathematician behind the constant e just defined, among a staggering number of other contributions – yet contemporaries described him as warm and good-natured rather than possessive of his own discoveries. He kept this good nature even after going completely blind in his later years, continuing to produce mathematics of great originality for another two decades – leaving behind so much unpublished work that it kept his assistants busy for four more decades after his death.*

5.3 Sequences and closed intervals

Before moving on to limits of functions, we pause to record two facts about sequences that live inside a closed interval $[a, b]$. Both are purely about sequences (they need nothing beyond the definition of convergence and the notion of a subsequence), and both will be essential tools later, specifically in Chapter 9.

The Bolzano-Weierstrass Theorem

Let $\{x_n\}_{n\in\mathbb{N}}$ be a bounded sequence (i.e., there exist real numbers $a < b$ such that $a < x_n < b$ for all $n \in \mathbb{N}$). Then $\{x_n\}_{n\in\mathbb{N}}$ has a convergent subsequence: there exists a strictly increasing sequence of indices $\{n_k\}_{k\in\mathbb{N}} \subseteq \mathbb{N}$ such that the subsequence $\{x_{n_k}\}_{k\in\mathbb{N}}$ converges.

Proof. We use a **bisection method** to construct a nested sequence of closed intervals, each containing infinitely many terms of $\{x_n\}_{n\in\mathbb{N}}$.

Construction. Set $[a_0, b_0] = [a, b]$; by hypothesis it contains infinitely many terms of the sequence. Given $[a_k, b_k]$ containing infinitely many terms, split it into its left and right halves. At least one of the two halves must still contain infinitely many terms (otherwise $[a_k, b_k]$ itself would contain only finitely many); call that half $[a_{k+1}, b_{k+1}]$ (if both halves qualify, pick either one, say the left one). By construction,

$$[a_0, b_0] \supseteq [a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots, \qquad b_k - a_k = \frac{b-a}{2^k}.$$

The endpoints converge. The sequence $\{a_k\}_{k\in\mathbb{N}}$ is monotone increasing (each interval is nested in the previous one) and bounded above by b, so by the Monotone Convergence Theorem (Theorem 5.1) it converges; call its limit $\bar{x}$. Likewise $\{b_k\}_{k\in\mathbb{N}}$ is monotone decreasing and bounded below by a, so it too converges. Since $2^k \geq k + 1$ for all $k \in \mathbb{N}$ (an easy induction), $2^k \to +\infty$, so for any $\varepsilon > 0$ we can choose k large enough that $\frac{b-a}{2^k} < \varepsilon$; hence $b_k - a_k \to 0$, and $\{b_k\}_{k\in\mathbb{N}}$ converges to the same limit $\bar{x}$.

Extracting the subsequence. Since each $[a_k, b_k]$ contains infinitely many terms of $\{x_n\}_{n\in\mathbb{N}}$, we can always choose an index beyond the ones used so far: pick n_0 with $x_{n_0} \in [a_0, b_0]$, and, having chosen $n_0 < n_1 < \cdots < n_{k-1}$, pick $n_k > n_{k-1}$ with $x_{n_k} \in [a_k, b_k]$ (possible precisely because $[a_k, b_k]$ contains infinitely many terms, so not all of them have index $\leq n_{k-1}$). This gives a strictly increasing sequence of indices $\{n_k\}_{k\in\mathbb{N}} \subseteq \mathbb{N}$ and a subsequence

$$x_{n_0}, x_{n_1}, x_{n_2}, x_{n_3}, \ldots, x_{n_k}, \ldots$$

The subsequence converges to $\bar{x}$. Since $x_{n_k} \in [a_k, b_k]$ and $\bar{x} \in [a_k, b_k]$ (being the common limit of a_k and b_k, with $a_k \leq \bar{x} \leq b_k$ for all k), we have

$$|x_{n_k} - \bar{x}| \leq b_k - a_k = \frac{b-a}{2^k} \xrightarrow[k\to\infty]{} 0.$$

Directly from the definition of the limit: given $\varepsilon > 0$, choose K with $\frac{b-a}{2^K} < \varepsilon$; then for all $k \geq K$, $|x_{n_k} - \bar{x}| \leq b_k - a_k \leq \frac{b-a}{2^K} < \varepsilon$. Hence $\lim_{k\to\infty} x_{n_k} = \bar{x}$, which completes the proof. (You will see this same bisection idea again, used to locate a root rather than a limit point, in the proof of Bolzano's Theorem in Chapter 9.) □

> **Bernard Bolzano (1781–1848)**
>
> *Bolzano was a Catholic priest and philosopher who held the chair of philosophy of religion in Prague, and who taught that war and militarism were a waste of society's resources, pressing instead for reform directed toward peace rather than armed conflict between nations. For refusing to recant these and other liberal views, he was formally removed from his professorship on Christmas Eve 1819. Barred from teaching, he spent much of the rest of his life – and much of his own income – supporting poorhouses, homes for the blind, and schools for the poor.*

Example 5.4: The sequence $a_n = (-1)^n$, $n \in \mathbb{N}$, has values

$$1, -1, 1, -1, 1, -1, \ldots, (-1)^n, (-1)^{n+1}, \ldots$$

This sequence is indeterminate. However, if we choose only the even indices we find that a_{n_k} (where $n_k = 2k$) is the subsequence $a_0, a_2, a_4, \cdots = 1, 1, 1, \ldots$ does have the trivial limit of 1. Similarly, the odd indices ($n_k = 2k + 1$) will give us a subsequence with the limit -1.

Remark: The above example demonstrates that a bounded sequence can have more than one convergent subsequence. The Bolzano-Weierstrass theorem tells us that there's *at least* one.

Proposition 5.3 (Convergent sequences in a closed interval)**:** Let $a < b$ and let $\{x_n\}_{n\in\mathbb{N}}$ be a sequence with $x_n \in [a, b]$ for all $n \in \mathbb{N}$. If $\{x_n\}_{n\in\mathbb{N}}$ converges, then

$$\lim_{n\to\infty} x_n \in [a, b].$$

Proof. Let $L = \lim_{n\to\infty} x_n$, and suppose for contradiction that $L > b$. Take $\varepsilon = \frac{L-b}{2} > 0$. By the definition of the limit, there exists $N \in \mathbb{N}$ such that for all $n > N$, $|x_n - L| < \varepsilon$, and in particular

$$x_n > L - \varepsilon = \frac{L+b}{2} > b,$$

contradicting $x_n \leq b$. Hence $L \leq b$. The argument that $L \geq a$ is entirely analogous (take $\varepsilon = \frac{a-L}{2}$ if $L < a$). Together, $L \in [a, b]$. □

We will use both of these results together in Chapter 9, to extract from a sequence confined to a closed interval $[a, b]$ a subsequence whose limit is guaranteed to still lie in $[a, b]$.